LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 1 Test (Set B): Chemical Reactions and Equations

Chapter / Topic: Chemical Reactions and Equations

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Write balanced chemical equations wherever asked.

SECTION A

6 marks

Choose the correct option.

Q1. Which of the following is NOT a chemical change? **[1]**

(a) Rusting of iron (b) Digestion of food
(c) Burning of paper (d) Melting of ice

Q2. In the reaction $2H_2S + SO_2 \rightarrow 3S + 2H_2O$, the substance that is oxidised is: **[1]**

(a) H_2S (b) SO_2
(c) S (d) H_2O

Q3. Which of these is an endothermic process? **[1]**

(a) Burning of natural gas
(b) Decomposition of calcium carbonate
(c) Respiration
(d) Reaction of quicklime with water

Q4. When lead nitrate is heated strongly, the products are: **[1]**

(a) lead oxide, nitrogen dioxide and oxygen
(b) lead and nitrogen
(c) lead oxide and nitrogen
(d) lead nitrite and oxygen

Q5. A balanced chemical equation has equal numbers of: **[1]**

(a) molecules on both sides
(b) atoms of each element on both sides
(c) compounds on both sides
(d) reactants and products

Q6. **Assertion (A):** Packets of chips are flushed with nitrogen gas.
Reason (R): Nitrogen is a highly reactive gas that kills bacteria. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. What is an oxidation reaction? Identify the substance oxidised in: $4Na + O_2 \rightarrow 2Na_2O$. **[2]**

OR

Write one balanced equation each for a decomposition reaction in which the energy is supplied as (a) heat and (b) light.

Q8. Why do we paint iron articles? Name the process that this prevents. **[2]**

SECTION C

6 marks

Q9. A shiny brown element X becomes black when heated in air. **[3]**

(i) Identify X and the black substance. **[1]**

(ii) Write the balanced equation. **[1]**

(iii) Name the type of reaction. **[1]**

Q10. Explain each of the following with one balanced equation: (i) displacement reaction, (ii) double displacement reaction, (iii) redox reaction. **[3]**

OR

Write balanced equations for what happens when: (a) zinc reacts with dilute sulphuric acid, (b) quicklime reacts with water, (c) electric current is passed through water.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) What is corrosion? What is rust, and what conditions does iron need to rust? Give two ways to prevent rusting. **[3]**

(ii) What is rancidity? Give two ways to prevent it. **[2]**

OR

(a) Balance these equations: (i) $Fe_2O_3 + Al \rightarrow Al_2O_3 + Fe$ (ii) $BaCl_2 + K_2SO_4 \rightarrow BaSO_4 + KCl$ (iii) $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$ [3]
(b) Why is it necessary to balance a chemical equation? [2]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. A student filled a plastic mug with water and added a few drops of dilute sulphuric acid. Two carbon electrodes were fixed in the mug and connected to a 6 V battery. Test tubes filled with water were inverted over the electrodes. After some time, gas collected in both test tubes, and the volume of gas in one tube was double that in the other. **[4]**

(i) Why were a few drops of dilute sulphuric acid added to the water? **[1]**

(ii) Which gas was collected in the larger volume? **[1]**

(iii) Write the balanced equation and name the type of reaction. How would you test the gas collected in the smaller volume? **[2]**

~ All the best! ~